

CHAPTER 1

SAFETY



CHAPTER 1

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SAFETY

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SECTION 1-1

SAFETY PRECAUTIONS

Peerless mixers are designed and built to operate as safely as possible. However, if the precautions and procedures explained in this manual are not followed at all times, your mixer, like any heavy equipment, has the potential to cause serious personal injury or death.

The safety systems and features on **Peerless** mixers are designed and installed to protect one mixer operator only. Using a helper to assist the operator in discharging dough is not considered safe and must never be done. *Never reach into the bowl when someone else is next to the mixer's operator panel.*

Allow only qualified personnel to operate or otherwise service the mixer. **Peerless** strongly recommends that every person involved with mixer operation or service be required to read and understand **Chapter 1, Safety**, of the manual. In addition, every person performing a specific task involving mixer operation or service should be required to read and understand the chapter(s) pertaining to that task.

The **Mandatory Reading** list below details persons and departments typically involved in mixer operation and service, along with specific chapters recommended for reading by those persons.

MANDATORY READING:

Mixer operators.	Chapter 1, Safety Chapter 4, Operation
Sanitation personnel.	Chapter 1, Safety Chapter 6, Sanitation
Maintenance personnel.	Chapter 1, Safety Chapter 4, Operation Chapter 5, Maintenance Chapter 6, Sanitation
Contractors.	Chapter 1, Safety Chapter 3, Installation
Supervisors.	Entire Manual

If any part of this manual is unclear, call the Service Department at **Peerless Machinery Corp.** for assistance. The following page contains a brief list of the most important safety precautions and procedures. They must be followed at all times. The page has been duplicated in Spanish, French and German. If you require another language, please call the Engineering Department at **Peerless Machinery Corp.**

SECTION 1-1

SAFETY PRECAUTIONS

⚠ DANGER

Running or coasting mixing arm can cause serious injury or death. Never reach hand or arm inside bowl while mixing arm is moving.

⚠ DANGER

Moving dough can cause serious injury. Never reach into bowl through optional **ingredient door** with dough or agitator moving.

⚠ DANGER

Running or coasting mixing arm can cause serious injury or death. Never place upper body or head inside mixing bowl without disconnecting and locking out power.

⚠ DANGER

Rotating sprockets and chains or belts and pulleys can cause serious injury or death. Never run mixer with end doors open. Never open the mixer end doors while mixer is running or coasting. Never run mixer without guards in place. Disconnect and lockout power before working on mixer.

⚠ DANGER

High electrical voltage is present in electrical enclosure. Only qualified personnel should open electrical enclosure door. Disconnect and lockout power before working on mixer.

⚠ DANGER

Having more than one mixer operator at the same time creates a hazardous situation which can result in serious injury or death. *Never reach into the bowl while someone else is next to the mixer's operator panel.*

SECCION 1-1

PRECAUCIONES DE SEGURIDAD

⚠ PELIGRO

La operación o marcha por inercia del brazo mezclador podría ocasionar heridas graves o aún la muerte. No ponga nunca la mano o el brazo dentro del tazón mientras el brazo mezclador se esté moviendo.

⚠ PELIGRO

La masa en movimiento podría ocasionar heridas graves. No ponga nunca la mano en el tazón a través la puerta opcional para ingredientes mientras la masa o el agitador se estén moviendo.

⚠ PELIGRO

La operación o marcha por inercia del brazo mezclador podría ocasionar heridas graves o aún la muerte. No ponga nunca el torso o la cabeza dentro del tazón de mezclar sin haber previamente desconectado y trabado la corriente.

⚠ PELIGRO

Las ruedas dentadas que giran así como cadenas o correas y poleas podrían ocasionar heridas graves o aún la muerte. No opere nunca la mezcladora con las puertas en los extremos abiertas. No abra nunca las puertas en los extremos de la mezcladora mientras la mezcladora esté en operación o en marcha por inercia. No opere nunca la mezcladora sin que los sistemas de seguridad estén en su lugar. Desconecte y trabe la corriente antes de efectuar trabajos en la mezcladora.

⚠ PELIGRO

Hay alto voltaje eléctrico en el recinto eléctrico. La puerta del recinto eléctrico debe ser abierta únicamente por personal calificado. Desconecte y trabe la corriente antes de efectuar trabajos en la mezcladora.

⚠ PELIGRO

La presencia de más de un operador del mezclador produce una situación de peligro que puede causar lesiones graves o la muerte. Por ningún motivo introduzca las manos dentro del recipiente mientras haya otra persona al lado del panel del operador del mezclador.

SECTION 1-1

PRECAUTIONS DE SECURITE

⚠ DANGER

L'opération ou tournage à vide du bras du mixeur pourrait causer des blessures graves ou la mort. Ne jamais mettre la main ou le bras dans le bol pendant que le bras du mixeur bouge.

⚠ DANGER

La pâte en mouvement pourrait causer des blessures graves. Ne jamais mettre la main dans le bol à travers la porte optionnelle d'ingrédients tandis que la pâte ou l'agitateur bougent.

⚠ DANGER

Le bras du mixeur en opération ou tournant à vide pourrait causer des blessures graves ou la mort. Ne jamais placer le torse ou la tête dans le bol du mixeur sans avoir préalablement débranché et bloqué le courant.

⚠ DANGER

Les pignons et chaînes tournants ou courroies et poulies pourraient causer des blessures graves ou la mort. Ne jamais opérer le mixeur pendant que les portes aux extrémités soient ouvertes. Ne jamais ouvrir les portes des extrêmes du mixeur pendant que le mixeur est en marche ou tourne à vide. Ne jamais opérer le mixeur sans que les sauvegardes soient en place. Débrancher et bloquer le courant avant d'effectuer des travaux sur le mixeur.

⚠ DANGER

Une électricité de haut voltage est présente dans le boîtier électrique. La porte du boîtier électrique ne devrait être ouverte que par le personnel qualifié. Débrancher et bloquer le courant avant de travailler sur le mixeur.

⚠ DANGER

La présence simultanée de plus d'un opérateur crée une situation dangereuse qui pourrait provoquer un accident sérieux voire fatal. Ne mettez jamais la main dans la cuve si quelqu'un se trouve près du tableau de contrôle du pétrin.

ABSCHNITT 1-1

SICHERHEITSMASSNAHMEN

⚠ GEFAHR

Ein laufender oder sich noch bewegender Teigknetearm kann schwere oder tödliche Verletzungen verursachen. Niemals mit der Hand oder dem Arm in den Kessel langen, solange sich der Teigknetearm bewegt.

⚠ GEFAHR

Sich bewegender Teig kann schwere Verletzungen verursachen. Niemals durch die wahlweise erhältliche Zutatentür in den Kessel langen, wenn sich der Teig oder der Teigknetearm bewegt.

⚠ GEFAHR

Ein laufender oder sich noch bewegender Teigknetearm kann schwere oder tödliche Verletzungen verursachen. Niemals den Oberkörper oder Kopf in den Kessel stecken, ohne vorher die Stromzufuhr zu unterbrechen und durch ein Schloß zu sichern.

⚠ GEFAHR

Sich drehende Kettenräder und Ketten bzw. Riemen und Riemenscheiben können schwere oder tödliche Verletzungen verursachen. Niemals den Mixer mit offenen Endtüren laufen lassen. Niemals die Endtüren des Mixers öffnen, solange der Mixer läuft oder sich noch bewegt. Den Mixer niemals laufen lassen, wenn nicht alle Schutzvorrichtungen angebracht sind. Vor Arbeiten am Mixer die Stromzufuhr unterbrechen und durch ein Schloß sichern.

⚠ GEFAHR

Hohe elektrische Spannung ist im elektrischen Gehäuse vorhanden. Nur qualifizierte Fachleute sollten die Tür zum elektrischen Gehäuse öffnen. Vor Arbeiten am Mixer die Stromzufuhr unterbrechen und durch ein Schloß sichern.

⚠ GEFAHR

Es besteht die Gefahr ernstlicher oder tödlicher Verletzungen, wenn die Teigrührmaschine von mehr als einer Arbeitskraft gehandhabt wird. Niemals in die Teigschale greifen, solange sich eine weitere Person in der Nähe der Bedientafel befindet.

SECTION 1-2

PERSONAL PROTECTIVE EQUIPMENT

We recommend the use of **personal protective equipment** by operators, maintenance, service and cleaning personnel, and other personnel working in the operating area of a **Peerless** mixer when conditions warrant. **Personal protective equipment** includes safety eye glasses, ear plugs with the specified noise reduction rating (NRR) for the job, and respirators with adequate filters.

Peerless mixers operate within **Occupational Safety and Health Administration (OSHA)** guidelines which state that a person can safely perform an operation for 8 hours under a decibel (dBA) rating of 90. In operation, a typical **Peerless** mixer produces a noise level of 88 dBA at the rear of the mixer and 80 dBA at the front; although these noise levels will vary depending on the environment the mixer is placed into. The average person can safely operate a **Peerless** mixer without hearing protection, but other equipment in the area may exceed **OSHA** noise levels making hearing protection necessary.

The addition or replacement of a mixer may alter ambient noise levels significantly. We recommend that noise monitoring be performed any time a change in production, process, equipment, or controls could increase noise exposures over the **OSHA** action level of 85 dBA.

Because a mixer operator works with materials that could splash or spread dust, we recommend that eye protection be worn while certain tasks are being performed. During the mixer's **Automatic Jog and Tilt** cycle, for example, pieces of discharging dough could separate from the batch and strike the operator. Wearing safety glasses helps guard against eye irritation or injury from loose-flying dough.

We highly recommend wearing eye protection (chemical splash goggles) during the mixer sanitation process. Splashing water and caustic cleaning chemicals from the high-pressure wash can cause irritation or injury to the eyes. It might also be necessary to wear chemical gloves, aprons, boots, and face shields during sanitation.

Depending on how ingredients are loaded into the mixer, we recommend that the operator wear a respirator for protection from airborne particulants. For example, if flour is hand loaded, wearing a respirator guards against throat and lung irritation from flour dust.

SECTION 1-3

ZERO ENERGY STATE PROCEDURES

Before servicing, inspecting or cleaning your **Peerless** mixer, remove all energy sources (Zero Energy State) from the mixer and lock out so they cannot accidentally be restored. This state is generally referred to as **LockOut/TagOut**. **Peerless Machinery Corp. (Peerless)** highly recommends this practice be followed at all times. In addition, **OSHA Standard 1910.147, The Control of Hazardous Energy**, requires that all companies have and enforce a written **LockOut/TagOut** program.

Mixer Energy Sources Chart 1-3-1 and the following example, **LockOut Procedure Example 1-3-2**, are intended to be used as guidelines for writing the lockout procedure for your **Peerless** mixer. You must establish, implement and enforce your own particular safety rules and procedures.

The type of energy source, its lockout location and the preferred method of locking out are listed below. In addition, **Example 1-3-2** shows an example form for the lockout of a **Peerless** mixer.

MIXER ENERGY SOURCES CHART 1-3-1

ENERGY SOURCE	LOCATION	LOCKOUT PROCEDURE
Electrical	Electrical Enclosure	Shut power off using disconnect handle . Attach lock to opening above handle.
Hydraulic	Left-hand end of mixer	Block bowl. Crack cylinder lines to relieve pressure.
Pneumatic (Air)	Left-hand end of mixer	Shut off air valve on filter regulator assembly. Place lock through air valve.
Gravity	Many large component parts of mixer including bowl	Tie off or block as needed.
Thermal	Refrigeration coolant	See lockout procedure by refrigeration system manufacturer.
Chemical	Refrigeration coolant	See lockout procedure by refrigeration system manufacturer.

SECTION 1-3**LOCKOUT PROCEDURE EXAMPLE 1-3-2**

TYPE: Tilt Bowl Mixer

POWER SOURCE: X **ELECTRICAL** X **HYDRAULIC** X **PNEUMATIC** X **GRAVITY**

POTENTIAL ENERGY SOURCE: Jacket coolant pressure (Freon, ammonia, glycol, water)

POWER SOURCES STORED: X **YES** **NO**

IF YES, DESCRIBE: Hydraulic: Energy and gravity are present in the bowl tilting functions. Place the bowl in the full-down position, against bowl stop block.

VOLTAGE: **PHASE:**

DISCONNECT LOCATION(S): Mixer remote starter enclosure, disconnect handle on outside front of starter enclosure.

TYPE OF LOCK-OUT NEEDED: X **LOCK** X **CLAMP** **VALVE** X **OTHER**

LOCK-OUT LOCATION(S): Electrical-starter enclosure, pneumatic-hand valve on air panel inside mixer enclosure.

LOCK-OUT PROCEDURES:

1. Push STOP button. Stop all motors.
2. Identify all power sources and potential energy sources.
3. If working on bowl tilt functions, place bowl in the full-down position. *Stored energy--source: gravity.
4. If working on pneumatics, block out all air operated features (i.e., sponge door, ingredient door, flour gate). *Stored energy--source: gravity.
5. Pull disconnect lever, located on the starter enclosure, to the OFF position (down).
6. Place lockout clamp through hasp on disconnect.
7. Place padlock and tag on lockout clamp. Lock and remove the key.
8. Push START button on mixer to test for power.
9. Turn valve on air to OFF position and lock out.
10. Notify all affected people.

SECTION 1-4

WARNING LABEL DETAILS

In this section, all warning labels used on **Peerless** mixers are reproduced. A detailed explanation is given for each label.

Labels consist of the following:

- 1) **The signal word** (**DANGER, WARNING, CAUTION**)
- 2) **The hazard** (Usually a picture)
- 3) **What to do** (To prevent the hazard)

The signal words used on this machine and their meanings are:



Red label. There is potential for an immediate hazard with a high likelihood of severe personal injury or death if the hazard is not prevented.



Orange label. There is potential for a hazard which could result in severe personal injury or death if the hazard is not prevented.



Yellow label. There is potential for a hazard which could result in minor injury, product, or property damage if the hazard is not prevented.

See **Figure 1-5-1** for the location of each label on your mixer.

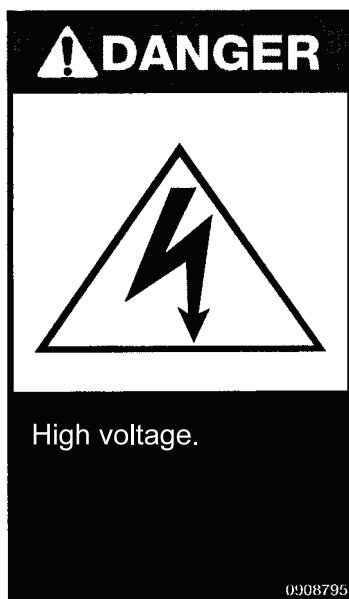
SECTION 1-4



This is a red **DANGER** label. It warns of a potential electrical shock. This shock can cause severe personal injury or death.

Before working on or cleaning this machine, the electrical power supply must be disconnected and locked out.

Figure 1-4-1 Label #0907310



This is a red **DANGER** label. It indicates the presence of high electrical voltage and warns of a potential electrical shock. This shock can cause severe personal injury or death.

Before working on or cleaning this machine, the electrical power supply must be disconnected and locked out.

Figure 1-4-2 Label #0908795

SECTION 1-4



Figure 1-4-3 Label #0908788

This is a red **DANGER** label. It warns of potential electrical shock if the cover is not secured. This shock can cause severe personal injury or death.

Before turning this machine on make sure the cover is installed and secured in place.

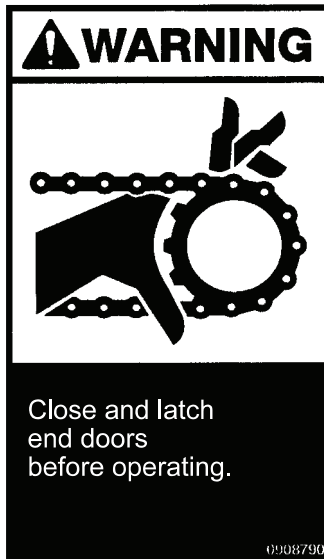


Figure 1-4-4 Label #0908789

This is an orange **WARNING** label. It warns of a potential hazard due to a rotating part. This hazard could result in severe personal injury.

Before turning power on to the machine, make sure the guard attached to this label or the guard in front of this label is installed and secured in place.

SECTION 1-4



This is an orange **WARNING** label. It warns of a potential hazard due to rotating sprockets and chain or belts. This hazard could result in severe personal injury or death.

Before turning power on to the machine, make sure the mixer end doors are closed and latched or that the mixer end doors are installed and secured in place.

Figure 1-4-5 Label #0908790



This is an orange **WARNING** label. It warns of a potential hazard due to rotating gears. This hazard could result in severe personal injury or death.

Before turning power on to the machine, make sure the mixer end doors are closed and latched or that the mixer end doors are installed and secured in place.

Figure 1-4-6 Label #0908791

SECTION 1-4



Figure 1-4-7 Label #0908792

This is an orange **WARNING** label. It warns of a potential pinching hazard due to the contact between two parts. This hazard could result in severe personal injury.

Before turning power on to the machine, make sure the mixer end doors are closed and latched or that the mixer end doors are installed and secured in place.



Figure 1-4-8 Label #0908793

This is an orange **WARNING** label. It warns of a potential hazard due to the movement of the agitator arm. This movement can be while jogging the mixer or by the coasting of the agitator arm. This movement can result in severe personal injury or death.

Do not reach inside the bowl when the agitator is moving. Never reach inside the bowl so far that your upper torso and head are in the path of the agitator arm without turning off and locking out power to the mixer.

SECTION 1-4



Figure 1-4-9 Label #0908794

This is an orange **WARNING** label. It warns of a potential pinching hazard due to the bowl closing against the bowl cover. This hazard can result in severe personal injury.

Keep hands away from the front edge of the bowl while the bowl is closing.

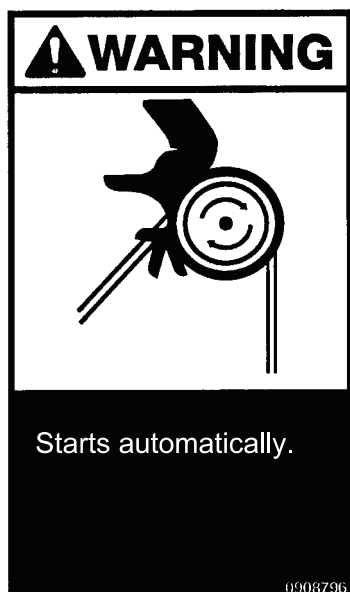


Figure 1-4-10 Label #0908796

This is an orange **WARNING** label. It warns of a potential pinching hazard due to the machine starting automatically. This hazard can result in severe personal injury.

Keep hands away from pulleys and gears.

SECTION 1-4



This is an orange **WARNING** label. It warns of a potential falling hazard due to climbing on the mixer. This hazard can result in severe personal injury.

Do not climb on mixer.

Figure 1-4-11 Label #0913927

SECTION 1-5

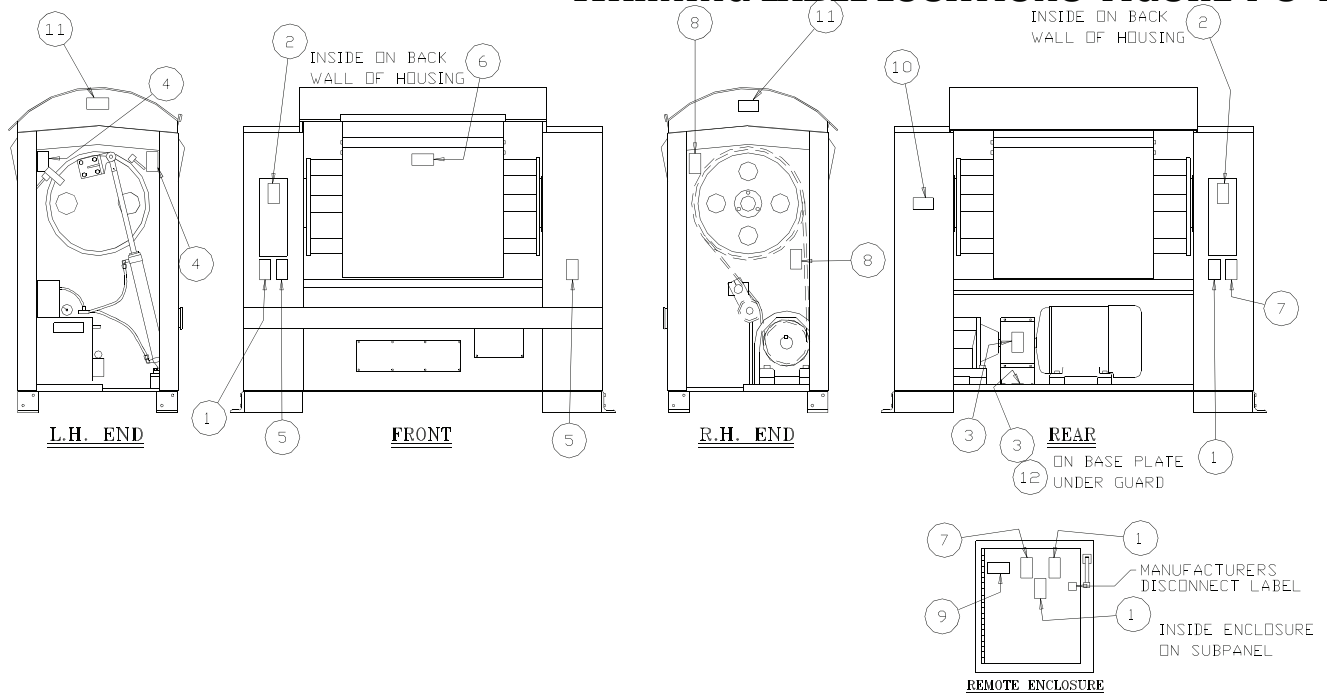
WARNING LABEL LOCATIONS

Figure 1-5-1 on the following page shows the warning label locations for your **Peerless** mixer. If at any time a label is removed or becomes unreadable, replace immediately. For new labels, call the **Peerless Machinery Corp.** Parts Department.

Peerless supplies replacement labels for its mixers *at no charge*.

SECTION 1-5

WARNING LABEL LOCATIONS FIGURE 1-5-1



ITEM	QTY	PART NO.	WARNING LABEL DESCRIPTION
1	4	907310	DANGER--Disconnect and Lockout Electrical Power Before Servicing or Cleaning
2	2	908788	DANGER--Install and Secure Cover Before Turning Power On
3	2	908789	WARNING--Do Not Operate Without Guard in Place
4	2	908792	WARNING--Close and Latch End Doors Before Operating
5	2	908793	DANGER--Do Not Reach Inside Bowl When Agitator Is Running or Coasting
6	1	908794	WARNING--Keep Hands Clear of Pinch Points
7	2	908795	DANGER--High Voltage
8	2	908790	WARNING--Close and Latch End Doors Before Operating
9	1	50A0213	Electrical Rating Name Plate
10	1	Call Peerless.	Jacket Information Tag (Mount across from Relief Valve)
11	2	913927	WARNING--Do Not Climb on Mixer Canopy
12	1	HS10-17-B2	Label Mounting Plate

SECTION 1-6

GENERAL SAFETY INFORMATION

This section details general basic safety information for machinery. Supervisors should use this information when setting up procedures for the operation, maintenance and cleaning of machinery including your **Peerless** mixer.

USER RESPONSIBILITY 1-6-1

The basic rules of safety in this section are intended as a guide for the safe operation of all machinery. All personnel who will be involved with any machine should become familiar with this information, and also with the safety information for the specific machine they are using.

It is the buyer's responsibility to make certain that the procedures outlined in this section and in the machine's manual are followed. It is also the buyer's responsibility to make sure that everyone involved with the operation of this machine understands the potential for hazard and injury.

Should any change be made to a machine, all procedures set for its safe operation must be reviewed and modified if needed. It is strongly recommended that you contact the original manufacturer to make certain the machine can be changed as desired and for suggestions about any changes in procedure.

SAFETY FIRST 1-6-2

Modern machines are designed and manufactured by following generally accepted safety standards. However, the safe operation of any machine depends on using safe operating, maintenance and servicing procedures under properly trained supervision.

For your protection and the protection of others, learn and always follow the safety rules outlined in your machine's manual. Observe all danger, warning and caution signs on the machine and act accordingly. Form safe working habits by reading all procedures and guidelines in this section and the machine's manual. Use good judgment when carrying out these procedures. Always preplan your work steps and ask questions when you are not sure of the steps.

SECTION 1-6**OPERATION SAFETY 1-6-3****OPERATION ZONE**

Prepare an operation zone around all installed machines. Define the operating space with a brightly painted guard rail or warning stripe. The operating space should be large enough so the operator has room to safely operate the machine and perform any additional duties required. Only the operator or other qualified personnel should be in the operating zone when the machine's power is on. Keep the operating zone clear of all unnecessary tools, equipment and other obstructions which may interfere with the safe operation of the machine.

SAFETY INSPECTION

Make the following safety inspections before starting the machine and after shutting it down. Also, refer to any additional pre-operation checkouts and instructions included in the machine's manual.

Before starting machine:

- ✓ Be absolutely sure all guards and safety devices are installed and in working order.
- ✓ Be sure all personnel are clear of the machine.
- ✓ Remove any materials, tools or other objects that do not belong in the operating zone.
- ✓ Make certain all indicating lights, horns, pressure gauges or other safety devices or indicators are in working order.

After shut down:

Make certain all pneumatic, hydraulic and electrical power is turned off.

OPERATION GUIDELINES

- ✓ Do not operate any machine until you have read and understand the machine's operating instructions and have become thoroughly familiar with the machine and its controls.
- ✓ Be sure all safety guards are in place before operating any machine. Safety guards and shields are on the machine for your protection and are to be removed only as required for repairs and maintenance by qualified maintenance personnel.
- ✓ Be sure the machine and all safety devices are fully operational. Never operate any machine if a component is faulty, damaged or while a safety device is disconnected or bypassed.
- ✓ Always wear safety glasses and other safety equipment and clothing as conditions require. Follow the guidelines in the machine's manual.
- ✓ Read and follow closely **DANGER**, **WARNING** and **CAUTION** labels located on the machine. **DANGER**, **WARNING** and **CAUTION** labels are placed on the machine where a hazardous condition may exist. *They are there for your protection and must not be removed or defaced.* Torn or worn labels should be replaced.
- ✓ Warn and remove all personnel not qualified or not required to be in the operating zone before the machine's power is turned on and operation begins.
- ✓ Always wear close-fitting clothing. Restrain any loose clothing or long hair to prevent it from getting caught in moving or rotating parts.
- ✓ Keep operating zone free of obstacles that could cause a person to trip or fall. Never sit or stand on anything that might cause you to fall against the machine.
- ✓ Know emergency shutdown procedures for the machine.
- ✓ Always operate the machine within specified limits for speed, pressure, temperatures, motor loads, etc.
- ✓ Always attend to the machine while in operation if constant attention is required. Keep alert and observe indicator lights and warnings that are displayed on the machine.
- ✓ Never place fingers, hands or any part of your body into the machine or near moving parts when electrical, hydraulic or air pressure are turned on.
- ✓ Disconnect all air, hydraulic and electrical power when the machine is not in use.

SECTION 1-6

CLEANING SAFETY 1-6-4

- ✓ Always clean up spills around the machine as soon as possible.
- ✓ Disconnect all pneumatic, hydraulic and electrical power from the machine before doing any cleaning procedures. Be sure the electrical power source is locked out using a padlock for which only you have a key. **OSHA 1910.147, The Control of Hazardous Energy (LockOut/TagOut)**.
- ✓ Use only methods and cleaning solutions specified in the machine's cleaning instructions. Do not use toxic and/or flammable solvents to clean the machine.
- ✓ Keep all electrical panel covers closed when washing the machine. Avoid spraying water or steam in the direction of an electrical panel.

SERVICE AND MAINTENANCE SAFETY 1-6-5

GENERAL SERVICE AND MAINTENANCE SAFETY

- ✓ Assign all service and maintenance to trained and qualified personnel only.
- ✓ Be sure no service or maintenance is being performed on the machine and that all personnel are clear before operating.
- ✓ Disconnect and lockout all pneumatic, hydraulic and electrical power to the machine unless those energies are required for the specific maintenance being performed. Use a padlock for which only you have the key. **OSHA 1910.147, The Control of Hazardous Energy (Lock-Out/TagOut)**.
- ✓ Carry out maintenance on a machine in motion only when it is absolutely necessary. When conditions require adjustments on a machine in motion, closely follow the maintenance instruction in the machine's manual and use extreme care.
- ✓ Check safety devices often and repair or replace if found faulty. Do not bypass any safety device to keep the machine running.

SECTION 1-6

- ✓ Check operation of all **EMERGENCY STOP** buttons often to make sure they function at all times. Repair or replace if found faulty.
- ✓ Always use the proper tool for the job.
- ✓ Disconnect any microprocessor or **Programable Logic Controller (PLC)** before doing any welding on a machine. A microprocessor or **PLC** used in a control circuit could be damaged by a voltage surge from the welding process.
- ✓ Use trained manufacturer service personnel to perform those maintenance or repair tasks that department personnel are not trained for. Know your maintenance department's limitations and plan work accordingly.
- ✓ Avoid maintenance personnel fatigue that commonly occurs from working long hours on downed machines or stress from projects with a tight deadline. Maintenance personnel fatigue causes accident and injury. Reduce risk by keeping enough qualified personnel on hand to operate and maintain the machines.

ELECTRICAL SAFETY

- ✓ Use only qualified electricians to perform electrical and electronic maintenance.
- ✓ Assume at all times that electrical power is turned ON and all circuits are "LIVE." Prevent accident or injury by taking a cautious approach.
- ✓ When removing electrical power from a circuit or machine, open the disconnect, or breaker, and lock in the OPEN position using a padlock for which only you have the key. Using only one key will prevent anyone else from accidentally turning on the electrical power while the machine is being worked on. (**OSHA 1910.147, The Control of Hazardous Energy LockOut/TagOut**). See **Section 1-3**.
- ✓ Make certain that the circuit is OPEN and no voltage is present by using the proper test meter. Test the meter often to make sure it is working.
- ✓ Replace fuses only when the electrical power is disconnected and locked out.
- ✓ Allow adequate time for capacitors to discharge.
- ✓ Reset starter overload relays (heaters) only when the electrical power is disconnected and locked out.
- ✓ If trouble-shooting on a live machine is necessary, take these special precautions:
 - Make sure your tools and body are not grounded.
 - Use only tools with insulated handles, specially designed for electricians.
 - Be alert, and avoid any outside distractions.
 - Do not attempt to service a live machine where water is present.
- ✓ Keep control panel doors closed unless servicing electrical equipment or wiring. After closing the panel door, make certain that the disconnect handle is operating properly.
- ✓ Close all covers on junction panels before turning on power to any machine.
- ✓ Before turning on power to a machine, make certain that all personnel are clear of it.

SECTION 1-6

HYDRAULIC SYSTEM SAFETY

- ✓ Be sure all covers, safety devices and indicators are in place and operating prior to turning on any hydraulic system.
- ✓ Repair all hydraulic system leaks and clean up all fluid before operating the system. Always clean hydraulic fluid from floors and machines before use. Standing hydraulic fluid is a fire hazard that can also cause someone to slip and fall into the machine.
- ✓ Operate the hydraulic system only at the hydraulic pressure settings specified in the machine's manual.
- ✓ Relieve all hydraulic pressure before performing maintenance.
- ✓ Avoid skin and eye contact with hydraulic fluid by wearing protective clothing, gloves and safety glasses.
- ✓ Do not attempt to perform maintenance or operate hydraulic equipment around an open flame. Never attempt to weld or flame cut on a hydraulic system unless the system has been thoroughly flushed of all hydraulic fluid.

PNEUMATIC SYSTEM SAFETY

- ✓ Always repair or replace leaking pneumatic (air) lines or components before operating a pneumatic system.
- ✓ Shut off and lock out shop air to the machine before doing any maintenance to the pneumatic system. Relieve all air pressure from the system.
- ✓ Always wear safety glasses when working on a pneumatic system.